

Manual Execution for Problems 1 and 2

Small graph 1

Source file: 'data/manual/directed_manual_1.txt'

Edges:

- 0 -> 1
- 0 -> 2
- 0 -> 3
- 1 -> 0
- 1 -> 3
- 1 -> 4
- 2 -> 1
- 2 -> 4
- 3 -> 2
- 4 -> 3

Problem 1: forward BFS from 0 to 4

Queue and parent map:

Step	Popped	Queue after step	New parents
---	---	---	---
0	-	0	'0: none'
1	0	1, 2, 3	'1: 0', '2: 0', '3: 0'
2	1	2, 3, 4	'4: 1'
3	2	3, 4	none
4	3	4	none
5	4	empty	stop

Reconstructed path: '0 -> 1 -> 4'

Lowest length: '2'

Problem 2: backward BFS from 0 to 4

Queue and next map:

Step	Popped	Queue after step	New next vertices
---	---	---	---
0	-	4	'4: none'
1	4	1, 2	'1: 4', '2: 4'
2	1	2, 0	'0: 1'
3	2	0, 3	'3: 2'
4	0	3	stop

Reconstructed path from start: '0 -> 1 -> 4'

Lowest length: '2'

Small graph 2

Source file: 'data/manual/directed_manual_2.txt'

Edges:

- 0 -> 1
- 0 -> 4
- 1 -> 2
- 1 -> 4
- 2 -> 0
- 2 -> 3
- 2 -> 4
- 3 -> 1
- 4 -> 2
- 4 -> 3

Problem 1: forward BFS from 3 to 0

Queue and parent map:

Step	Popped	Queue after step	New parents
---	---	---	---
0	-	3	'3: none'
1	3	1	'1: 3'
2	1	2, 4	'2: 1', '4: 1'
3	2	4, 0	'0: 2'
4	4	0	none
5	0	empty	stop

Reconstructed path: '3 -> 1 -> 2 -> 0'

Lowest length: '3'

Problem 2: backward BFS from 3 to 0

Queue and next map:

Step	Popped	Queue after step	New next vertices
---	---	---	---
0	-	0	'0: none'
1	0	2	'2: 0'
2	2	1, 4	'1: 2', '4: 2'
3	1	4, 3	'3: 1'
4	4	3, 0	none
5	3	0	stop

Reconstructed path from start: '3 -> 1 -> 2 -> 0'

Lowest length: '3'

Large graph results

The official datasets were run from the files in 'data/large'.

graph1k

Forward BFS:

- '1 -> 100': length '6', path '1 -> 5 -> 487 -> 175 -> 699 -> 624 -> 100'
- '100 -> 1': length '5', path '100 -> 416 -> 354 -> 865 -> 109 -> 1'

Backward BFS:

- '1 -> 100': length '6', path '1 -> 5 -> 487 -> 175 -> 699 -> 624 -> 100'
- '100 -> 1': length '5', path '100 -> 416 -> 354 -> 865 -> 109 -> 1'

graph10k

Forward BFS:

- '1 -> 100': length '8', path '1 -> 3300 -> 2607 -> 523 -> 6311 -> 5359 -> 9794 -> 5173 -> 100'
- '100 -> 1': length '7', path '100 -> 2398 -> 3054 -> 5232 -> 8217 -> 2478 -> 7151 -> 1'

Backward BFS:

- '1 -> 100': length '8', path '1 -> 7317 -> 4118 -> 2404 -> 690 -> 1494 -> 739 -> 4722 -> 100'
- '100 -> 1': length '7', path '100 -> 5568 -> 2781 -> 1451 -> 4997 -> 528 -> 4260 -> 1'

graph100k

Forward BFS:

- '1 -> 100': length '8', path '1 -> 17024 -> 27471 -> 14969 -> 3075 -> 4156 -> 32753 -> 14973 -> 100'
- '100 -> 1': length '8', path '100 -> 44340 -> 54527 -> 6606 -> 53263 -> 95930 -> 98655 -> 58288 -> 1'

Backward BFS:

- '1 -> 100': length '8', path '1 -> 17024 -> 27471 -> 14969 -> 3075 -> 4156 -> 32753 -> 14973 -> 100'
- '100 -> 1': length '8', path '100 -> 44340 -> 54527 -> 6606 -> 53263 -> 95930 -> 98655 -> 58288 -> 1'

Note: on 'graph10k', the forward and backward traversals return different paths for the same query, but the lengths are the same. That is valid because the graph has multiple shortest paths.